

LightGCN:
Simplifying and Powering Graph Convolution
Network for Recommendation

-ACM 2020-

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1. Introduction

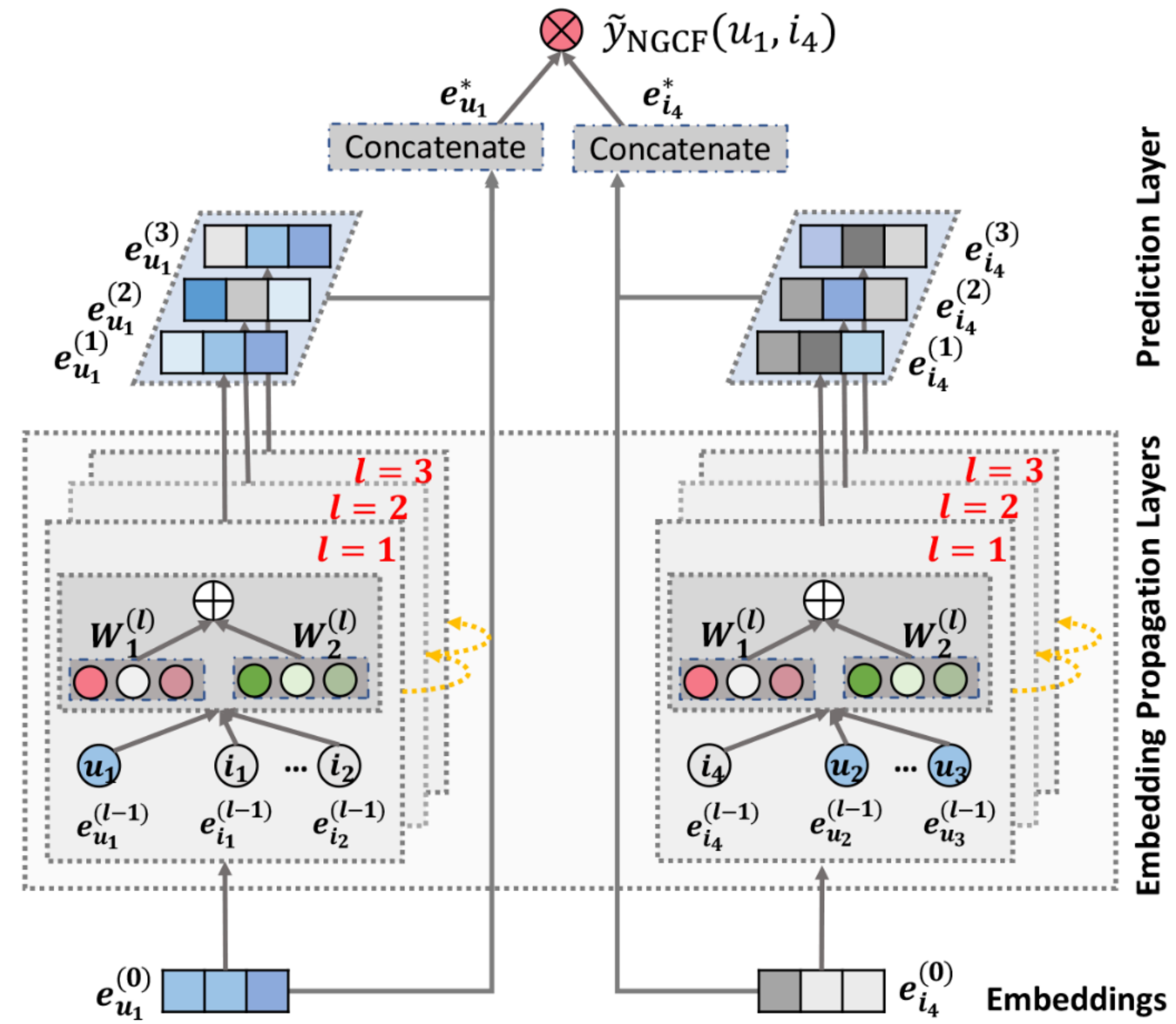
2. Model

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Introduction

NGCN Architecture



Introduction

NGCN

$\mathbf{e}_u^{(k)}$: k번째 layer 전파 후 user embedding

$\mathbf{e}_i^{(k)}$: k번째 layer 전파 후 item embedding

$\mathbf{W}$: 각 레이어에서 특징 변환을 수행하기 위한 학습 가능한 가중치 행렬

σ : 비선형 활성화 함수

$$\mathbf{e}_u^{(k+1)} = \sigma \left(\mathbf{W}_1 \mathbf{e}_u^{(k)} + \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} (\mathbf{W}_1 \mathbf{e}_i^{(k)} + \mathbf{W}_2 (\mathbf{e}_i^{(k)} \odot \mathbf{e}_u^{(k)})) \right),$$

$$\mathbf{e}_i^{(k+1)} = \sigma \left(\mathbf{W}_1 \mathbf{e}_i^{(k)} + \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} (\mathbf{W}_1 \mathbf{e}_u^{(k)} + \mathbf{W}_2 (\mathbf{e}_u^{(k)} \odot \mathbf{e}_i^{(k)})) \right),$$

Introduction

NGCN

$\mathbf{e}_u^{(k)}$: k번째 layer 전파 후 user embedding

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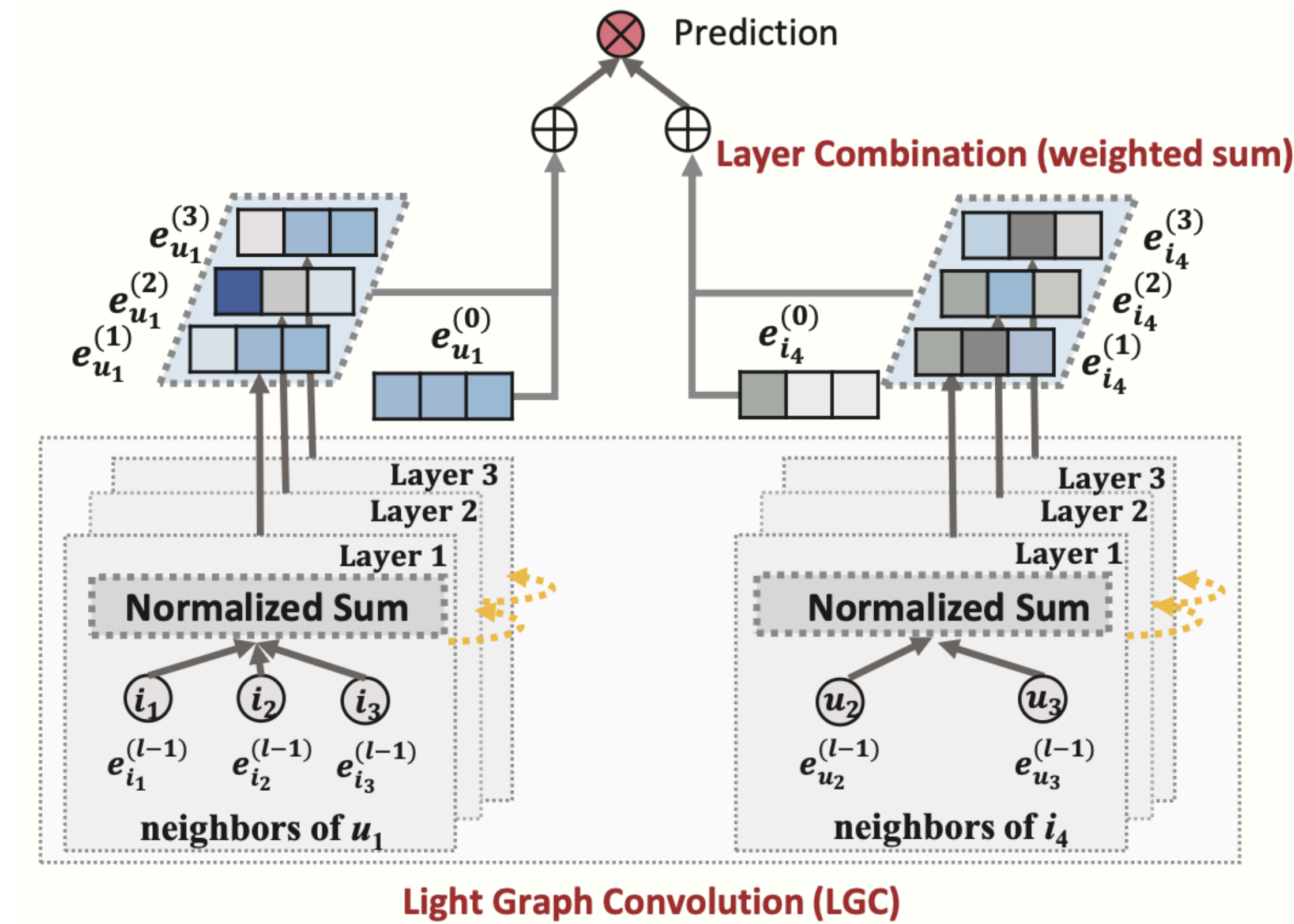
가중치 행렬, 비선형함수는
필요없다고 주장

$$\mathbf{e}_u^{(k+1)} = \sigma \left(\mathbf{W}_1 \mathbf{e}_u^{(k)} + \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} (\mathbf{W}_1 \mathbf{e}_i^{(k)} + \mathbf{W}_2 (\mathbf{e}_i^{(k)} \odot \mathbf{e}_u^{(k)})) \right),$$

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Model

LightGCN Architecture



Model

LightGCN

$$\mathbf{e}_u^{(k+1)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|} \sqrt{|\mathcal{N}_i|}} \mathbf{e}_i^{(k)},$$
$$\mathbf{e}_i^{(k+1)} = \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i|} \sqrt{|\mathcal{N}_u|}} \mathbf{e}_u^{(k)}.$$

LightGCN

$$\mathbf{e}_u^{(l)} = \text{LeakyReLU} \left(\mathbf{m}_{u \leftarrow u}^{(l)} + \sum_{i \in \mathcal{N}_u} \mathbf{m}_{u \leftarrow i}^{(l)} \right),$$

NGCN

Model

LightGCN

NGCN과 달리 Self-connection 이 없는 이유

- NGCN의 최종 임베딩 : Concat
- LightGCN 최종 임베딩 : 가중평균

$$\mathbf{e}_u = \sum_{k=0}^K \alpha_k \mathbf{e}_u^{(k)}; \quad \mathbf{e}_i = \sum_{k=0}^K \alpha_k \mathbf{e}_i^{(k)},$$

Model

LightGCN

a_k의 경우 학습가능하게 또는 수동 튜닝 가능

논문에서는 a_k : 1/(k+1)로 진행

- Over-smoothing 방지
- Self-connection 기능 수행

$$\mathbf{e}_u = \sum_{k=0}^K \alpha_k \mathbf{e}_u^{(k)}; \quad \mathbf{e}_i = \sum_{k=0}^K \alpha_k \mathbf{e}_i^{(k)},$$

Prediction

Score Prediction

- 앞서 가중평균한 User, Item임베딩들을 내적하여 Score 예측값 도출

$$\hat{y}_{ui} = \mathbf{e}_u^T \mathbf{e}_i,$$

Prediction

Object Loss

- BPR Loss 사용
- BPR Loss란?
 - 상호작용한 아이템(Positive)은 상호작용하지 않은 아이템(Negative)보다 예측점수가 높아야한다

u : User

i : u User가 관측한 Item(pos)

j : u User가 관측하지 않은 Item(neg)

σ : 시그모이드 함수(0~1 확률로 변환)

Θ : 러닝 파라미터 (W , item/user Embeddings)

λ : L2 정규화

$$Loss = \sum_{(u,i,j) \in O} -\ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|\Theta\|_2^2$$

Prediction

Train

학습에 L2정규화로만으로도 **overfitting**을 방지할 수 있으므로 Dropout을 사용하지 않는다

u : User

i : u User가 관측한 Item(pos)

j : u User가 관측하지 않은 Item(neg)

σ : 시그모이드 함수(0~1 확률로 변환)

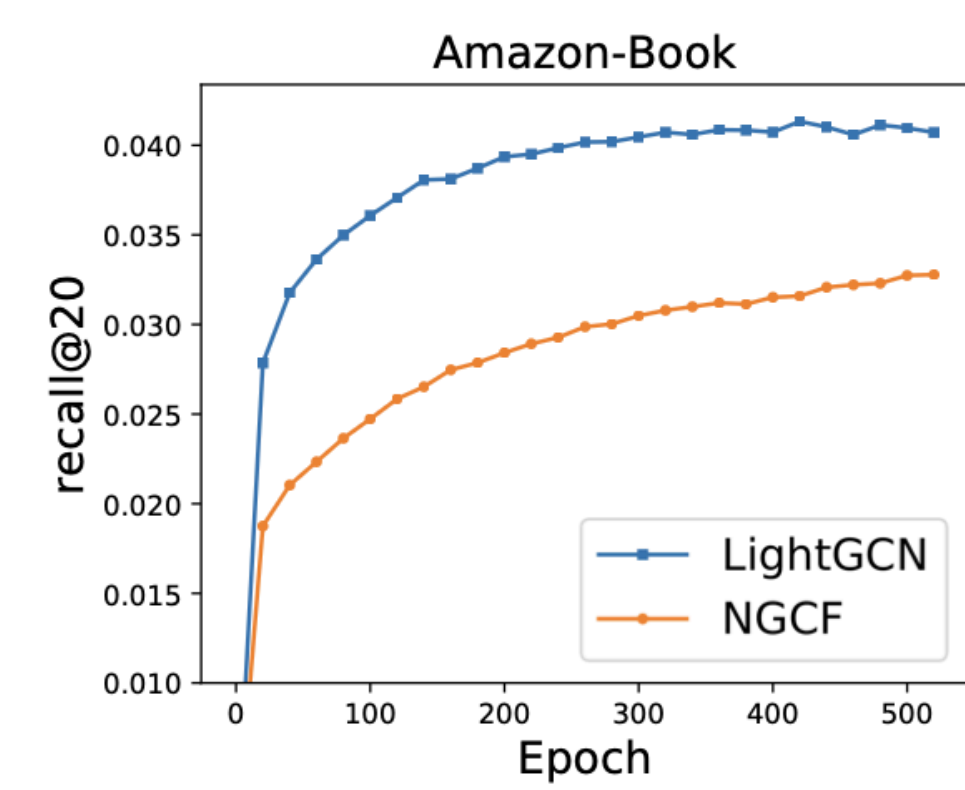
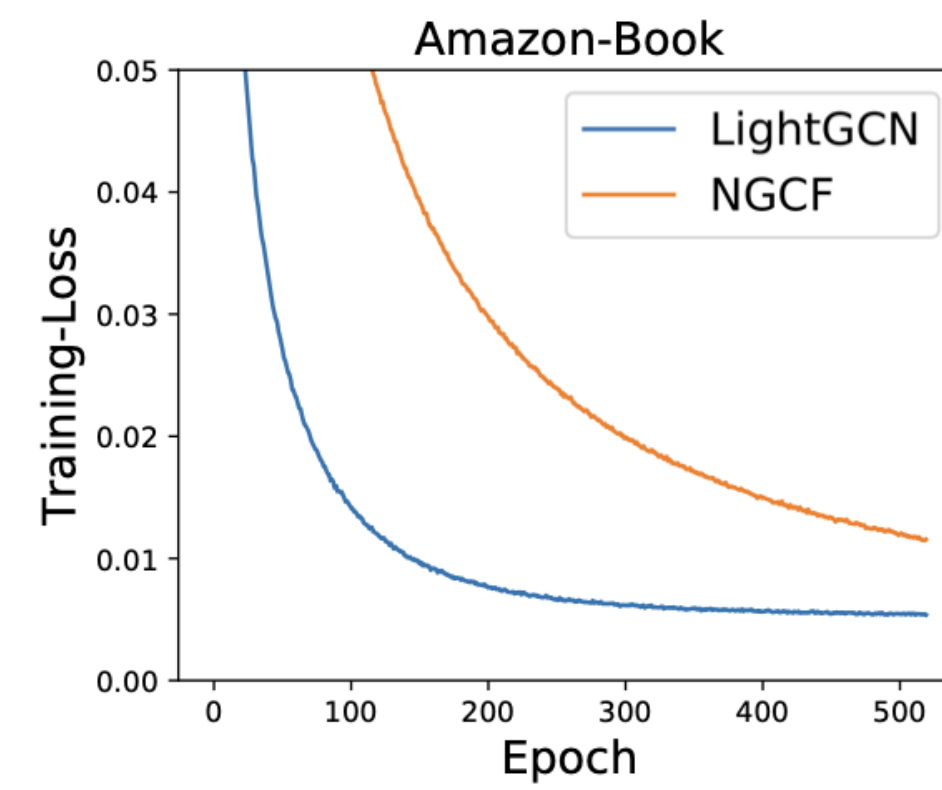
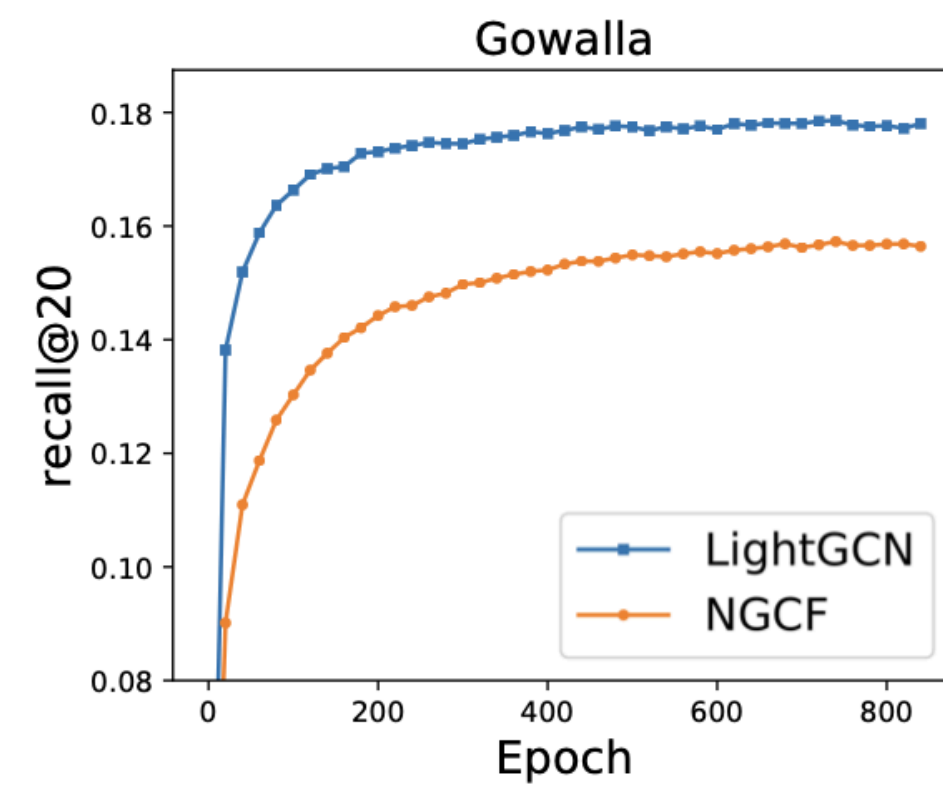
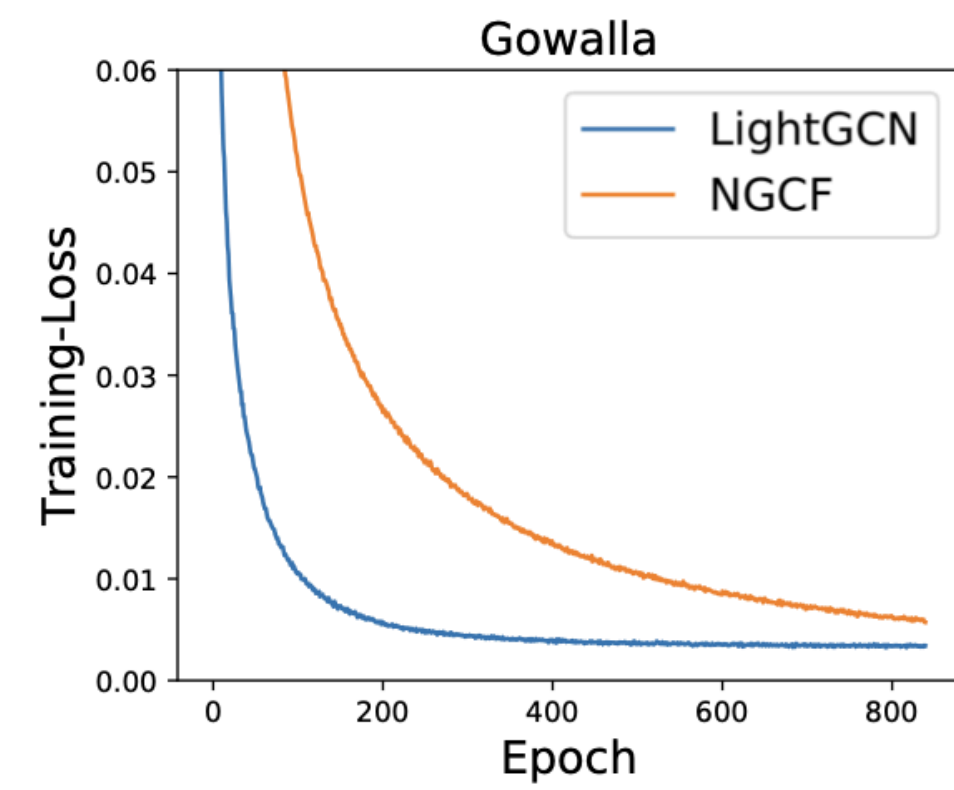
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Experiments

LightGCN vs NGCF



Experiments

LightGCN vs NGCF

Dataset		Gowalla		Yelp2018		Amazon-Book	
Layer #	Method	recall	ndcg	recall	ndcg	recall	ndcg
1 Layer	NGCF	0.1556	0.1315	0.0543	0.0442	0.0313	0.0241
	LightGCN	0.1755(+12.79%)	0.1492(+13.46%)	0.0631(+16.20%)	0.0515(+16.51%)	0.0384(+22.68%)	0.0298(+23.65%)
2 Layers	NGCF	0.1547	0.1307	0.0566	0.0465	0.0330	0.0254
	LightGCN	0.1777(+14.84%)	0.1524(+16.60%)	0.0622(+9.89%)	0.0504(+8.38%)	0.0411(+24.54%)	0.0315(+24.02%)
3 Layers	NGCF	0.1569	0.1327	0.0579	0.0477	0.0337	0.0261
	LightGCN	0.1823(+16.19%)	0.1555(+17.18%)	0.0639(+10.38%)	0.0525(+10.06%)	0.0410(+21.66%)	0.0318(+21.84%)
4 Layers	NGCF	0.1570	0.1327	0.0566	0.0461	0.0344	0.0263
	LightGCN	0.1830(+16.56%)	0.1550(+16.80%)	0.0649(+14.58%)	0.0530(+15.02%)	0.0406(+17.92%)	0.0313(+18.92%)

Experiments

LightGCN vs LightGCN-single

